

Lab6

July 25, 2025

[]:

1 Lab 6: Visualization of Multi-Dimensional Data

Description:

- Numeric features: math score, reading score, writing score.
- Categorical features: gender, race/ethnicity, parental level of education, lunch, test preparation course.
- Using 'gender' as the main category and create an 'average score' as a fourth numeric feature.

```
[1]: # Imports
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.decomposition import PCA
from sklearn.manifold import TSNE
import numpy as np

# Load dataset (upload StudentsPerformance.csv to Colab or use the path if
# already present)
df = pd.read_csv('StudentsPerformance.csv')

# Create average score as a new numeric feature
df['average score'] = df[['math score', 'reading score', 'writing score']].
    mean(axis=1)
df.head()
```

```
[1]:  gender  race/ethnicity  parental level of education      lunch \
0  female      group B      bachelor's degree      standard
1  female      group C      some college      standard
2  female      group B      master's degree      standard
3   male      group A      associate's degree  free/reduced
4   male      group C      some college      standard

test preparation course  math score  reading score  writing score \
0              none          72          72          74
1      completed          69          90          88
```

2	none	90	95	93
3	none	47	57	44
4	none	76	78	75

	average score
0	72.666667
1	82.333333
2	92.666667
3	49.333333
4	76.333333

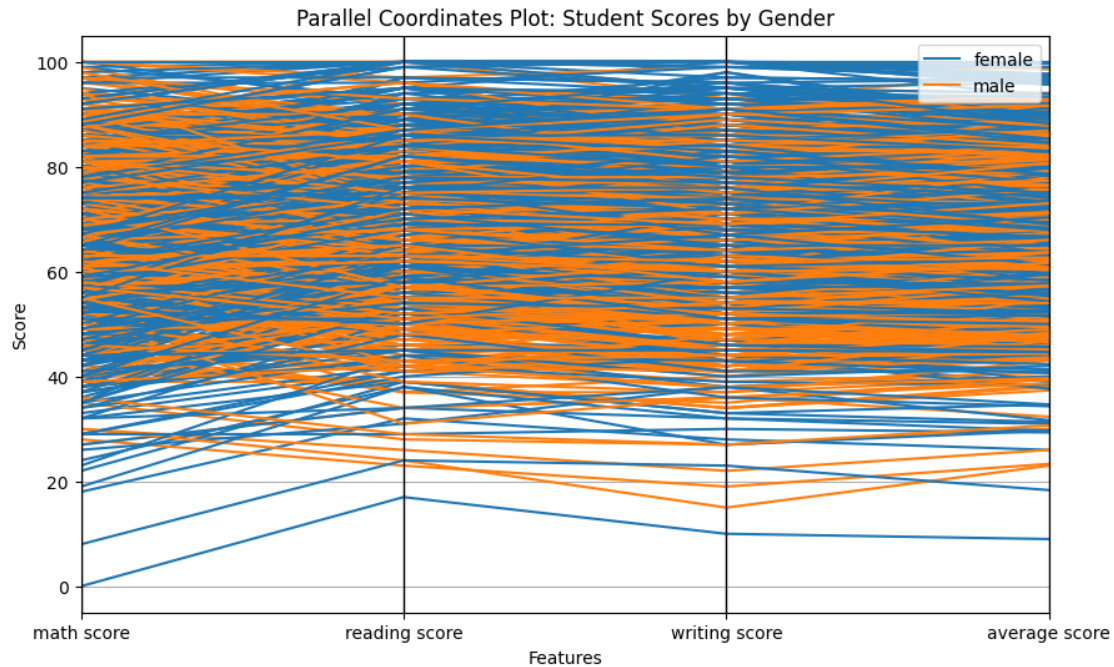
1.1 1. Parallel Coordinates Plot

We'll visualize four numeric features (math, reading, writing, average score) colored by gender.

Parallel coordinates help reveal multivariate patterns, such as how groups differ across several features at once, and can highlight clusters or outliers.

```
[2]: from pandas.plotting import parallel_coordinates

plt.figure(figsize=(10,6))
parallel_coordinates(
    df[['gender', 'math score', 'reading score', 'writing score', 'average_
    ↪score']],
    'gender',
    color=['#1f77b4', '#ff7f0e']
)
plt.title('Parallel Coordinates Plot: Student Scores by Gender')
plt.xlabel('Features')
plt.ylabel('Score')
plt.show()
```



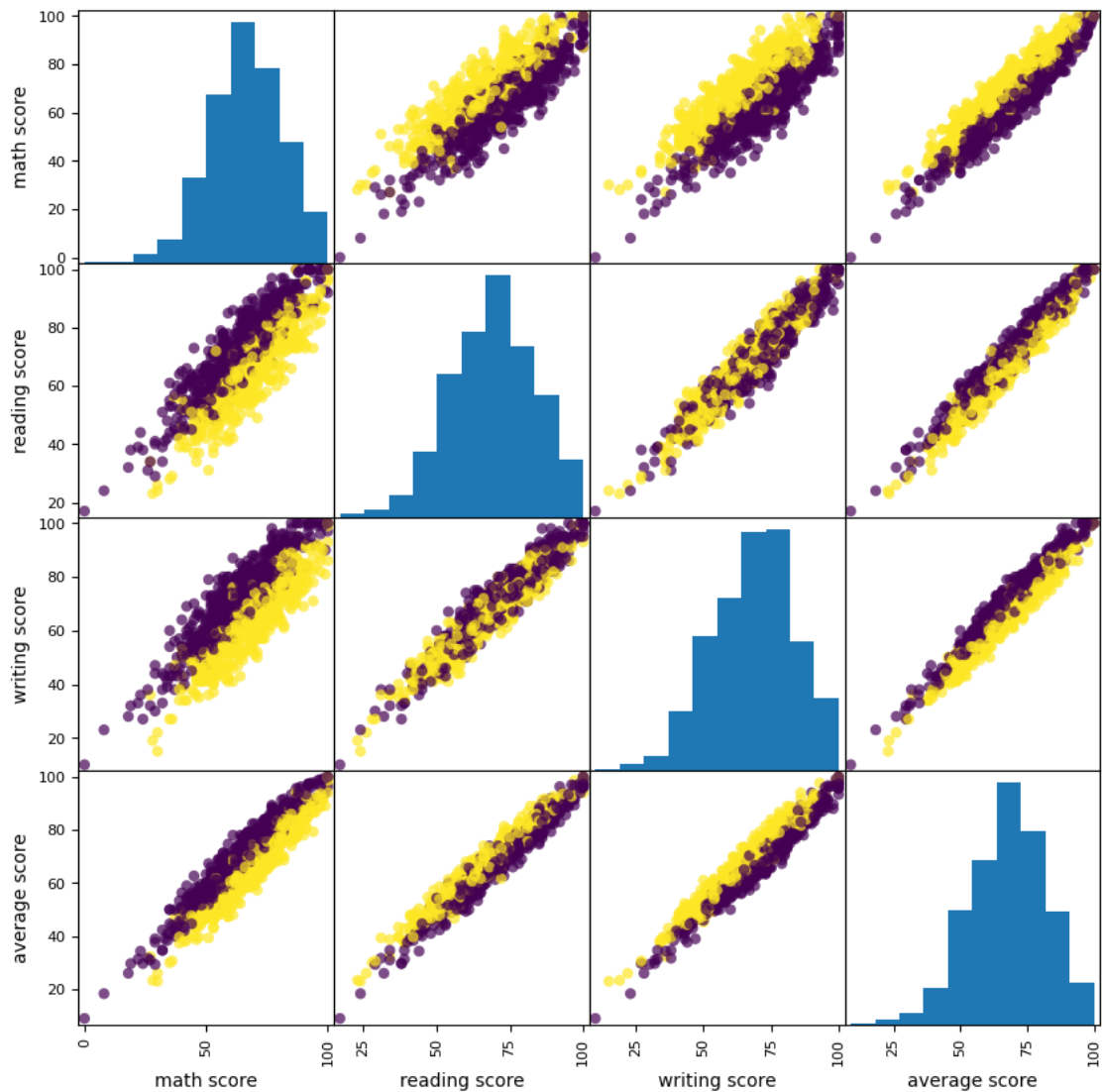
Observation: - Female students tend to have higher reading and writing scores, while male students have a slight edge in math. - Parallel coordinates make it easy to spot these trends across all features at once.

1.2 2. Scatter Matrix (Pandas)

A scatter matrix shows pairwise relationships between numeric features, colored by gender.

```
[3]: pd.plotting.scatter_matrix(
    df[['math score', 'reading score', 'writing score', 'average score']],
    figsize=(10,10),
    diagonal='hist',
    alpha=0.7,
    c=df['gender'].map({'female': 0, 'male': 1}),
    marker='o'
)
plt.suptitle('Scatter Matrix of Student Scores')
plt.show()
```

Scatter Matrix of Student Scores



Observation: - Strong positive relationship between reading and writing scores. - Math score shows more overlap between genders, indicating less separation.

1.3 3. Seaborn Pairplot

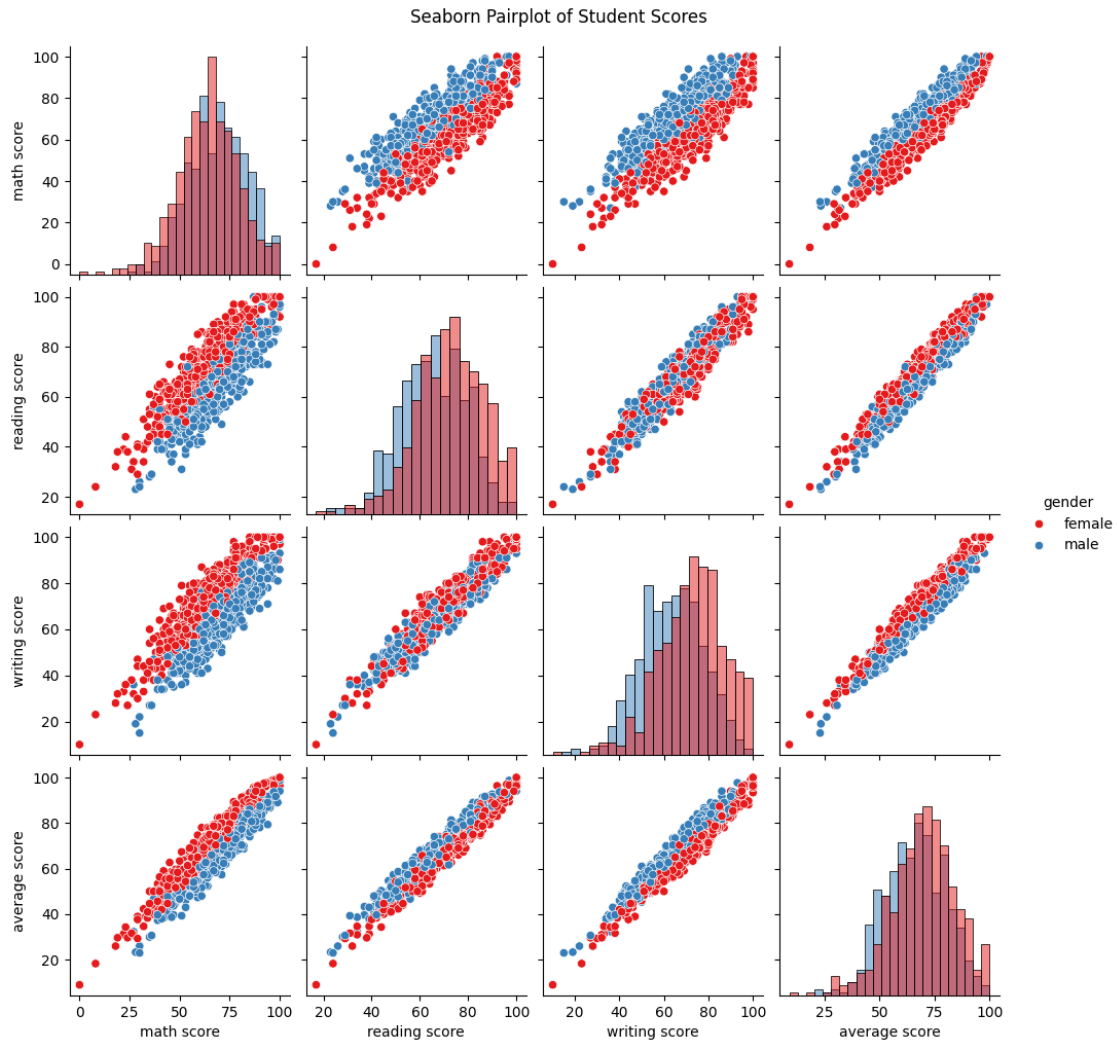
A pairplot provides a grid of scatterplots and histograms, colored by gender.

```
[4]: sns.pairplot(  
    df,  
    vars=['math score', 'reading score', 'writing score', 'average score'],
```

```

    hue='gender',
    palette='Set1',
    diag_kind='hist'
)
plt.suptitle('Seaborn Pairplot of Student Scores', y=1.02)
plt.show()

```



Observation: - Clear clusters for reading and writing scores by gender. - Some outliers visible, especially in math scores.

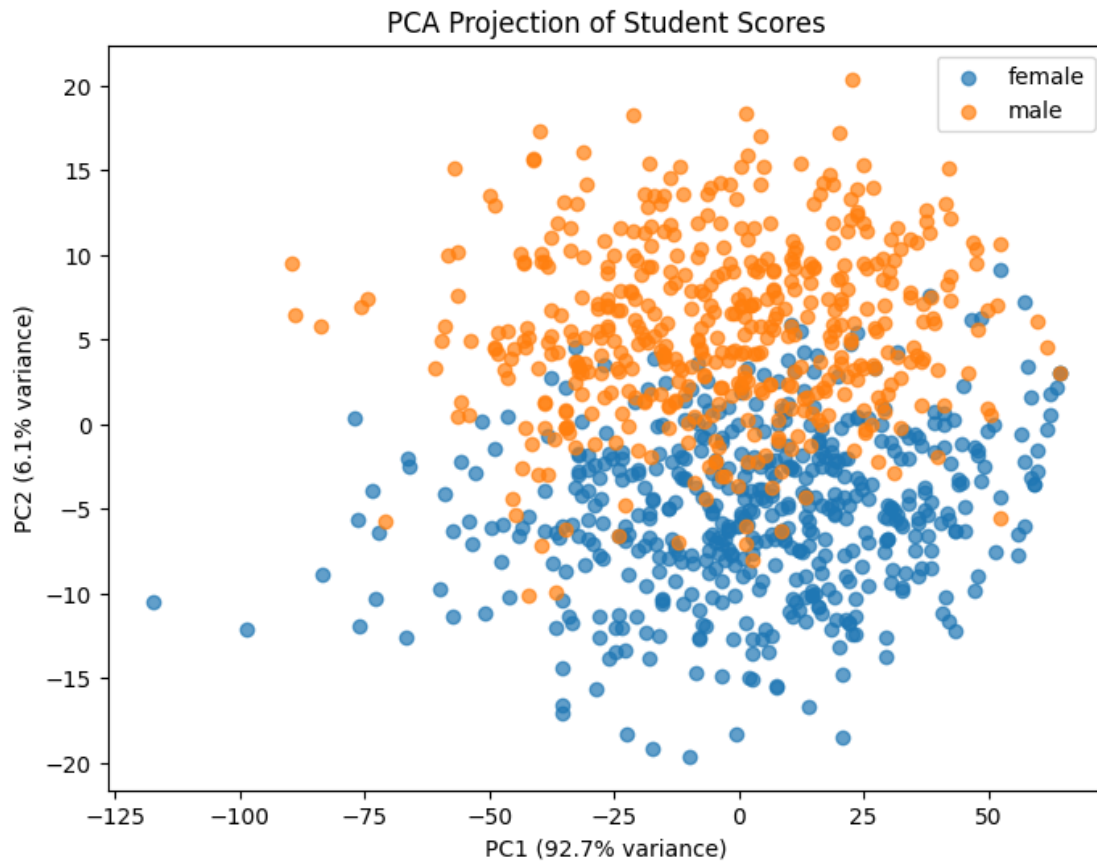
1.4 4. PCA Projection

We'll use PCA to reduce the four numeric features to two principal components and plot them, colored by gender.

```
[5]: features = ['math score', 'reading score', 'writing score', 'average score']
X = df[features].values
y = df['gender'].map({'female': 0, 'male': 1}).values

pca = PCA(n_components=2)
X_pca = pca.fit_transform(X)
explained = pca.explained_variance_ratio_ * 100

plt.figure(figsize=(8,6))
for i, label, color in zip([0,1], ['female', 'male'], ['#1f77b4', '#ff7f0e']):
    plt.scatter(X_pca[y==i,0], X_pca[y==i,1], label=label, alpha=0.7, c=color)
plt.xlabel(f'PC1 ({explained[0]:.1f}% variance)')
plt.ylabel(f'PC2 ({explained[1]:.1f}% variance)')
plt.title('PCA Projection of Student Scores')
plt.legend()
plt.show()
```



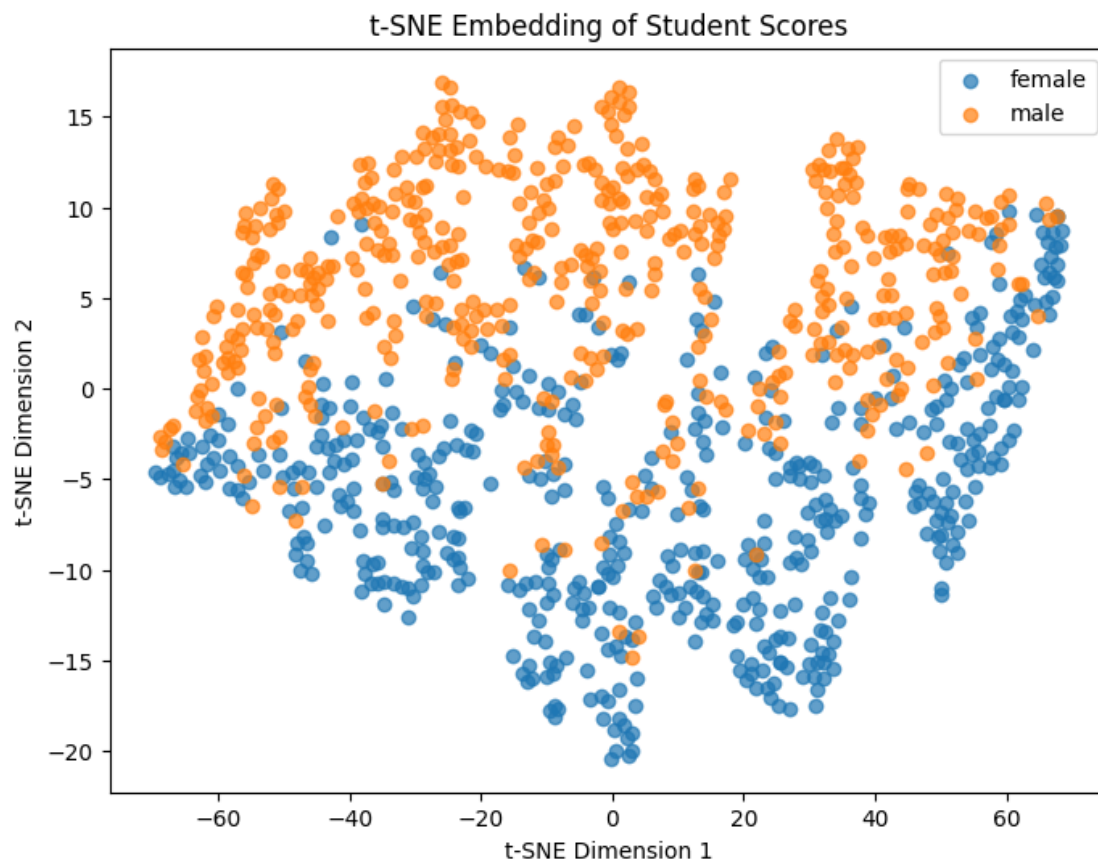
Observation: - PCA separates the genders moderately well, especially along PC2. - Most variance is explained by PC1, which combines all score types.

1.5 5. t-SNE Embedding

We'll use t-SNE to embed the four numeric features into two dimensions and plot, colored by gender.

```
[6]: tsne = TSNE(n_components=2, random_state=42, learning_rate=100)
X_tsne = tsne.fit_transform(X)

plt.figure(figsize=(8,6))
for i, label, color in zip([0,1], ['female', 'male'], ['#1f77b4', '#ff7f0e']):
    plt.scatter(X_tsne[y==i,0], X_tsne[y==i,1], label=label, alpha=0.7, c=color)
plt.xlabel('t-SNE Dimension 1')
plt.ylabel('t-SNE Dimension 2')
plt.title('t-SNE Embedding of Student Scores')
plt.legend()
plt.show()
```



Observation: - t-SNE forms tighter clusters by gender than PCA, especially for female students.
- t-SNE better preserves local groupings, while PCA explains more global variance.